

A Method of LFSR Seed Generation for Scan-Based BIST Using Constrained ATPG

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Abstract—This paper proposes a method of LFSR seed generation for scan-based BIST of VLSI circuits. The proposed method targets static faults and utilizes a commercial ATPG tool. Experimental results show that the proposed method can improve fault coverage and reduce the number of seeds compared with the conventional seed generation.

Keywords—Scan-based BIST; reseeding of LFSR; seed generation; constrained ATPG;

I. INTRODUCTION

Testing of a digital system is an experiment in which the system is exercised and its resulting response is analyzed to ascertain whether it behaved correctly. Testing approaches are divided into external testing and built-in self-test (BIST)[1]. In external testing the test equipment is external to the system under test; that is, test patterns are applied and the responses are then evaluated by the external tester. Although this achieves high fault coverage, using external tester is costly. As digital systems grow more complex it is more difficult to test, it becomes increasingly attractive to build self-testing aids into the unit under test. BIST is a design technique in which parts of a circuit are used to test the circuit itself. For very large scale integration (VLSI) circuits, BIST is important to reduce cost of manufacturing test and to enable in-field test after shipment[2]. Usually, BIST is used with scan design which is the most widely used design-for-testability (DFT) method[3]. A model of scan-based BIST is shown in Figure 1. A circuit with scan-based BIST consists of a sequential circuit with DFT, which is the circuit under test (marked as CUT in the figure), whose flip-flops are formed as a shift register (called as a scan path), a test pattern generator (TPG) and a response analyzer (RA).

A linear feedback shift register (LFSR) is usually used as a TPG. In general, there exist random-pattern-resistant faults (RPRFs) which are difficult to be detected by pseudo-random patterns generated by an LFSR in a circuit[4]. For such a circuit with RPRFs, reseeding[5][6][7] of the LFSR is effective to achieve high fault coverage. Some patterns are generated from an LFSR with some initial value (called as a *seed*) and then a new seed is set into the LFSR

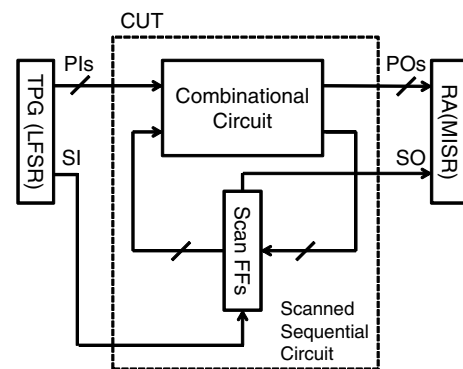


Figure 1. A model of scan-based BIST.

(called as *reseeding*) so as to detect some RPRF. So far, the following two types of LFSR seed generation methods have been proposed: random seed generation[6] and deterministic seed generation[5][7]. For the former type, some seeds are randomly generated by embedded circuitry. For the latter type, test patterns detecting RPRFs are generated using an automatic test pattern generation (ATPG) tool and each generated test pattern is converted into a seed. However, this method does not always succeed the conversion and may lose fault coverage. Furthermore there is a room for reducing the number of seeds converted. With our best knowledge, the proposed method is the first approach to generate seeds directly using an ATPG tool. The proposed method enables to evaluate the ability of the embedded BIST circuit in terms of fault detection.

In this paper, we propose a method to generate LFSR seeds to improve fault coverage and reduce the number of seeds for scan-based BIST. For a given circuit with scan-based BIST, the proposed method first creates a seed generation model of the circuit. The seed generation model consists of the combinational part of the circuit and the XOR network representing the logic function of the LFSR seed variables for each scan flip-flop or primary input. LFSR seeds for RPRFs are then generated as test patterns of the

seed generation model using an ATPG tool.

In this paper, we also evaluate the effectiveness of the method by experiments using ITC'99 benchmark circuits. The experimental results show that the proposed method can generate seeds with shorter time and can achieve higher fault coverage than the conventional method. In addition, the proposed method can reduce the numbers of the seeds to up to 36%.

This paper is organized as follows. Section II explains the scan-based BIST model shown in Figure 1 and RPRFs. Section III introduces a conventional seed generation method. Section IV propose a new seed generation method. Section V presents experimental results of the proposed seed generation method and the conventional one. Finally, Section VI presents our conclusions.

II. TARGET BIST MODEL

A. LFSR based BIST

A BIST model targeted in this paper is shown in Figure 1. In general, an LFSR and a multiple input signature register (MISR) are used as a TPG and an RA, respectively. An LFSR can generate pseudo-random patterns except the all zero pattern. The generated pattern is applied to the CUT directly from PIs and by way of scan-FFs.

Under normal operation, the primary inputs and the primary outputs of the CUT are directly controlled and observed, respectively, from outside the circuit. During test mode, test patterns are applied to the PIs and are shifted in to the scan-FFs from the LFSR, the responses of the circuit to these patterns are compacted through the MISR, and the signature of the MISR is compared to good one.

The scan path can be the plural. In this paper, we consider the TPG part but not the RA part.

B. Random pattern resistant fault

Consider an AND gate with n -inputs. When applying unbiased random inputs, the probability of justifying 1 on the output is $1/2^n$. A 0 on any input makes it impossible to test the stuck-at-0 fault on the output. Application of test pattern generated by an LFSR is difficult to detect such a fault. Such a fault is called a random pattern resistant fault (RPRF).

III. CONVENTIONAL METHOD

Given a set of deterministic test cubes¹, a conventional way is to compute a set of seeds that when expanded by the LFSR will produce the deterministic test cubes. For each test cube, a seed can be computed by solving a system of linear equations based on the feedback polynomial of the LFSR[5].

For example, consider an example circuit with primary inputs, PI_0 and PI_1 , and scan flip-flops (FFs), SFF_0 , SFF_2

¹A test vector in which bits unassigned by ATPG are left as don't cares.

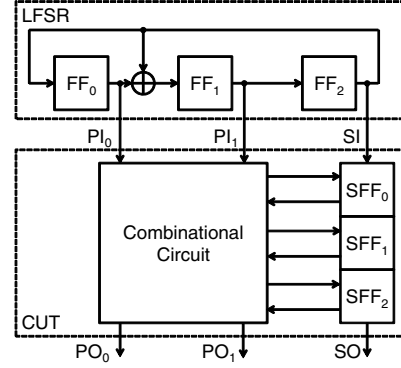


Figure 2. An example of CUT and LFSR.

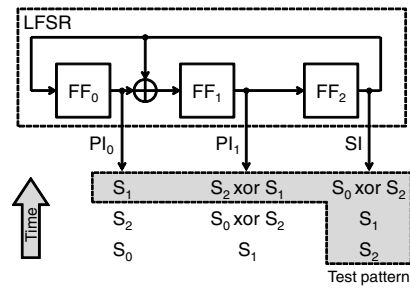


Figure 3. A test pattern expanded by LFSR from a seed.

and SFF_3 , shown in Figure 2. A test pattern is generated from the LFSR by applying three cycles under scan shift mode where the three cycles corresponds to the number of FFs on the scan path. Suppose that the seed of the LFSR is $(FF_0, FF_1, FF_2) = (S_0, S_1, S_2)$. Each coordinate of the expanded pattern can be represented by the functions of the seed. In this case, $(PI_0, PI_1, SFF_0, SFF_2, SFF_3)$ is $(S_1, S_2 \text{ xor } S_1, S_0 \text{ xor } S_2, S_1, S_2)$ as shown in Figure 3.

In the conventional method, a test cube $(PI_0, PI_1, SFF_0, SFF_2, SFF_3) = (T_0, T_1, T_2, T_3, T_4)$ is generated and converted to a seed by solving the following system of linear equations: $(T_0, T_1, T_2, T_3, T_4) = (S_1, S_2 \text{ xor } S_1, S_0 \text{ xor } S_2, S_1, S_2)$.

Although this method is widely used so far, there is a drawback that the conversion does not always succeed. It means that it may induce fault coverage loss. A flowchart of the conventional method for LFSR seed generation is shown in Figure 4. An ATPG tool may not generate a test cube, which can be converted into a seeds, for a detectable fault of a CUT under the BIST model because the ATPG tool does not know the conversion. That is the reason why the coverage loss occurs in the seed conversion process.

For example, a test cube $(PI_0, PI_1, SFF_0, SFF_2, SFF_3) = (1, 1, 0, 0, 1)$ is generated by using an ATPG tool. This pattern can not be converted to a seed because the system of linear equations $(1, 1, 0, 0, 1) = (S_1, S_2 \text{ xor } S_1, S_0 \text{ xor } S_2, S_1, S_2)$ is

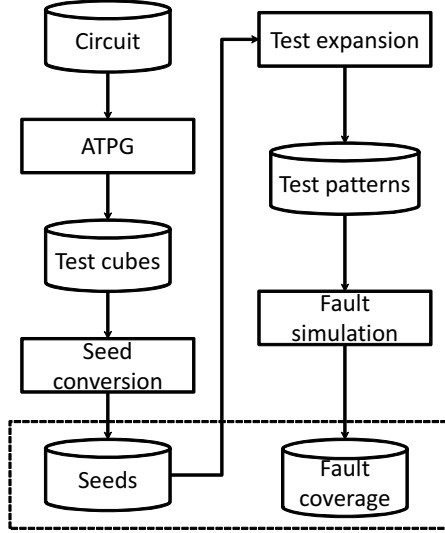


Figure 4. A flowchart of the conventional seed generation method.

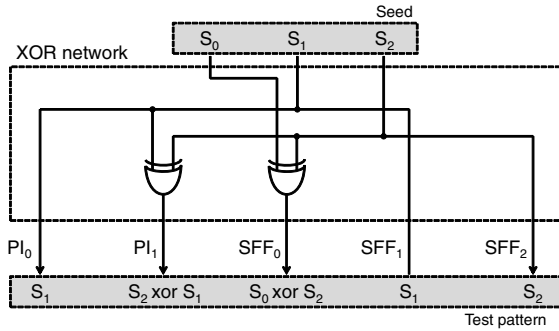


Figure 5. An XOR network representing three time frame expansion of the LFSR.

unsolvable.

We need to emphasize that we must run fault simulation for test patterns expanded from the obtained seeds if the quality of the test patterns needs to be evaluated.

IV. PROPOSED METHOD

To remove the drawback of the conventional method, we propose a method to generate LFSR seeds directly using an ATPG tool. The main idea of the method is to use an XOR network, which is obtained by time frame expansion of the LFSR, as an constraint for ATPG. The number of time frames expanded is equal to the length of the longest scan path in the CUT.

Consider the example circuit of Figure 2 again. The LFSR can be expanded with three time frames from a seed as Figure 3. For testing the CUT, only the shaded part is meaningful. The part can be represented by a combinational circuit as shown in Figure 5. We call such a combinational

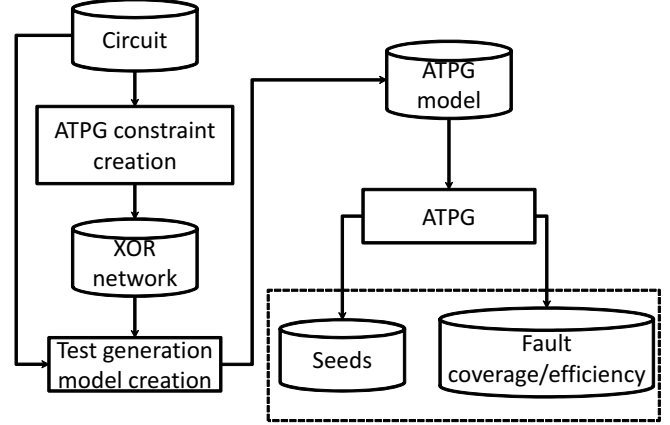


Figure 6. A flowchart of the proposed seed generation method.

circuit an *XOR network*.

In the proposed method, we employ an ATPG tool for generating seeds. The XOR network represents the relation between a seed and a test pattern for the primary inputs and scan FFs of the CUT. To inform the relation to the ATPG tool, we create a *seed generation model* which consists of the combinational part of the CUT and the XOR network.

For example, the XOR network (Figure 5) of the example circuit (Figure 2) and the combinational part of the CUT are connected as shown in Figure 7. By targeting a fault in the combinational part, an ATPG tool can generate a seed that when expanded by the LFSR will detect the fault or can prove that the fault is unstable.

The proposed method has advantages that ATPG tool can generate all the LFSR seeds for detectable faults by the BIST model and calculate fault coverage and fault efficiency. So it is not necessary to perform fault simulation for test patterns expanded from the obtained seeds. The overall flow of the proposed method is shown in Figure 6.

Notice that the seed generation model is only used during seed generation or no modification is made for the physical design of the circuit.

V. EXPERIMENTAL RESULTS

We evaluate the effectiveness of our proposed method by experiments. In the experiments, we used ITC'99 benchmark circuits[8]. The characteristics for each circuit is shown in Table I. Columns “#PIs”, “#POs”, “#Gates”, “#FFs” show the numbers of primary inputs, primary outputs, gates, FFs, respectively. The largest circuit among the benchmark circuits is b19. We inserted multiple scan paths for each circuit. The number of scan paths for each circuit is shown in Table II. For b19, we prepared three circuits with the different numbers of scan paths.

We first extracted random pattern resistant faults for each circuit. We used an LFSR of 100 bits for every circuit.

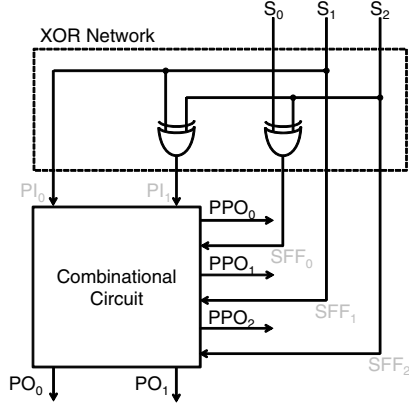


Figure 7. A seed generation model composed of the combinational part and XOR network

Table I
THE CHARACTERISTICS OF ITC'99 BENCHMARK CIRCUITS.

Circuit	#PIs	#POs	#Gates	#FFs
b14	32	54	8,567	245
b15	36	70	7,922	449
b17	37	97	27,852	1,415
b18	37	23	94,249	3,320
b19	24	30	190,213	6,642
b20	32	22	17,158	490
b21	32	22	17,482	490
b22	32	22	25,460	735

The feedback polynomial for the LFSR is selected from [9]. We applied 10,000 pseudo-random patterns generated from some seed for stuck-at faults using the fault simulator of TetraMAX (Synopsys). We assumed that the remaining faults are RPRFs which will be targeted for seed generation. Table III shows the number of RPRFs for each circuit. Column “Total # Faults”, “%FC” and “#RPRFs” show the numbers of faults, fault coverage in % and the number of RPRFs, respectively.

We implemented the conventional method by a SAT solver. We employed MiniSat[10] as a SAT solver. Table IV shows the results of the conventional method. Columns “%FC”, “%FE”, “CPU(sec.)” and “#Seeds” show fault coverage obtained using the pseudo-random patterns expanded from the generated seeds, fault efficiency², CPU time required for ATPG and conversion in second and the number of seeds converted by the method, respectively. The numbers reported in the parenthesis show the results without conversion. In the experiments, TetraMax was also used as an ATPG tool. The CPU used in the experiments was Xeon X5675 3.06GHz (Intel) with 40GB memory.

The results for our proposed method are shown in Table

²The ratio of the number of faults detected or proved untestable to the total number of faults.

Table II
THE CHARACTERISTICS OF SCAN PATHS FOR ITC'99 BENCHMARK CIRCUITS.

Circuit	# Scan Paths	Length
b14	3	82
b15	5	90
b17	24	59
b18	3	1,107
b19_scan6	6	1,107
b19_scan13	13	511
b19_scan22	22	302
b20	5	98
b21	17	30
b22	5	147

Table III
RANDOM PATTERN RESISTANT FAULTS OF ITC'99 BENCHMARK CIRCUITS.

Circuit	Total # Faults	%FC	#RPRFs
b14	60,080	87.52	7,469
b15	55,586	79.42	11,389
b17	197,992	64.83	69,638
b18	687,576	71.98	192,571
b19_scan6	1,385,822	70.17	413,297
b19_scan13	1,385,822	70.35	410,826
b19_scan22	1,385,822	70.03	415,265
b20	120,882	91.04	10,751
b21	123,220	87.48	15,377
b22	179,614	90.60	16,779

V. From the results, our proposed method can achieve higher fault coverage for every circuit with comparable CPU time compared with the conventional method. The overall comparison is shown in Table VI. Columns “Speed Up(%)”, “%FC Improv.”, and “# Seed Red.” show the ratio of the CPU time of the conventional method to that of the proposed method, fault coverage improvement from that of the conventional method to that of the proposed method, and the ratio of the number of seeds reduced by the proposed method to the number of seeds generated by the conventional method, respectively. For example, consider the result of b21. The improvement in the fault coverage is from 3.15% to 76.33%. The improvement is 73.18%. For all the circuits expect for b14, b20, b21 and b22, the proposed method reduced the numbers of seeds compared with the conventional method. For example, consider the result of b19_scan6, 63.04% of seeds are reduced by the proposed method compared with the conventional method. For b14, b20, b21 and b22, the number of seeds increases but the fault coverage increases dramatically like b21.

VI. CONCLUSION

In this paper, we have proposed an LFSR seed generation method for scan-based BIST. In contrast to the conventional method, LFSR seeds are directly generated by using our seed generation model with a commercial ATPG tool. In our

Table IV
RESULTS OF THE CONVENTIONAL SEED GENERATION.

Circuit	%FC	%FE	CPU(sec.)	#Seeds
b14	36.16 (96.09)	— (99.87)	0.80 (0.40)	206 (945)
b15	59.69 (85.09)	— (91.76)	2.70 (2.13)	733 (1,295)
b17	47.40 (92.75)	— (96.09)	13.01 (9.61)	2,336 (8,330)
b18	95.22 (97.90)	— (99.30)	34.51 (21.64)	22,541 (23,583)
b19_scan6	91.36 (96.45)	— (97.99)	100.43 (78.31)	46,772 (50,537)
b19_scan13	92.26 (96.42)	— (97.97)	103.53 (84.46)	46,872 (49,991)
b19_scan22	86.29 (96.48)	— (98.01)	142.65 (123.67)	43,784 (50,438)
b20	39.35 (94.74)	— (99.86)	1.31 (0.86)	349 (1,375)
b21	3.15 (94.75)	— (98.86)	1.73 (0.96)	56 (1,711)
b22	58.06 (95.78)	— (99.87)	1.94 (1.13)	841 (1,949)

Table V
RESULTS OF THE PROPOSED SEED GENERATION.

Circuit	%FC	%FE	CPU(sec.)	#Seeds
b14	96.06	99.87	0.45	660
b15	74.75	90.82	1.83	447
b17	81.63	94.71	7.53	2,121
b18	97.07	98.43	20.25	8,896
b19_scan6	94.73	96.43	86.80	17,285
b19_scan13	95.98	97.74	63.07	17,941
b19_scan22	93.32	97.74	65.32	17,594
b20	90.30	99.37	0.87	1,012
b21	76.33	97.72	3.77	941
b22	95.21	99.13	1.30	1,455

experiments, the proposed method can achieve higher fault coverage with smaller numbers of seeds for ITC'99 benchmark circuits compared with the conventional method. Although the coverage loss occurs in the conventional method, it does not occur in the proposed method. Furthermore, detectability of all the faults with a BIST model can be evaluated using the proposed method.

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Table VI
THE OVERALL COMPARISON.

Circuit	Speed Up(%)	%FC Improv.	# Seeds Red.(%)
b14	177.78	59.90	—
b15	147.54	15.06	39.02
b17	172.78	34.23	9.20
b18	170.41	1.85	60.53
b19_scan6	115.70	3.37	63.04
b19_scan13	164.15	3.72	61.72
b19_scan22	218.39	7.03	59.82
b20	150.57	50.95	—
b21	45.89	73.18	—
b22	149.23	37.15	—

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